

Project initialization and planning phase

Date	12 July 2026
Team Id	XXXXX
Project Name	Opti Crop: Smart Agricultural Production Optimization engine
Maximum marks	3 marks

Define problem Statement (customer Problem statements Template):

Smart Agricultural Production Optimization Engine addresses this challenge by using machine learning to analyze soil and environmental parameters and recommend the most suitable crop for cultivation. The system provides accurate, data-driven crop recommendations, helping farmers improve productivity, maximize profits, optimize resource utilization, and promote sustainable agricultural practices.

Example:

I am	I'm trying to	But	Because	Which makes me feel
A farmer who wants to improve crop production.	Find the most suitable crop for my land using soil and climate data.	I don't have accurate information or expert knowledge.	Soil nutrients, climate and environmental conditions vary and are complex to analyze.	Uncertain and worried about low yield and financial loss.

Problem Statement (PS)	I am (Customer)	I'm trying to	BUT	Because	Which makes me feel
PS-1	A farmer	Select the most suitable crop for my land.	I don't know which crop is best for my soil and weather conditions .	I have limited knowledge of soil nutrients and changing climatic conditions.	Confused and worried about crop yield and income.
PS-2	An agricultural researcher	Analyze crop and environmental data accurately.	It is difficult to identify useful patterns from large agricultural datasets.	Manual analysis is timeconsuming and complex.	Frustrated and less productive.
PS-3	An agricultural officer	Recommend suitable crops to farmers.	I don't have a reliable system to provide accurate crop recommendations.	Crop selection depends on multiple environmental factors.	Concerned about giving incorrect recommendations.